Component Diagram Justification

Project: ATM System

1. Introduction

The component diagram represents the main components of the ATM system and shows how they communicate with each other. It follows a layered architecture and separates the user interface, authentication, transaction processing, bank communication, database storage, and ATM hardware.

2. Components Used

The diagram contains six main components:

1. **ATM UI:** Allows customers to enter their PIN, select transactions, and view account information.
2. **Authentication Service:** Validates the customer's PIN and handles authentication requests.
3. **Transaction Processor:** Processes withdrawals, deposits, balance inquiries, and fund transfers.
4. **Bank Server API:** Communicates with the bank's backend system to verify accounts and process transactions.
5. **Database:** Stores account details and transaction records.
6. **ATM Hardware Adapter:** Connects the software with devices such as the card reader, keypad, cash dispenser, and receipt printer.

3. Interfaces Used

The diagram uses provided and required interfaces to show communication between components.

- **Authentication Interface:** Allows the ATM UI to request customer authentication.
- **Transaction Interface:** Allows the ATM UI to send transaction requests to the Transaction Processor.
- **Bank API Interface:** Allows the Transaction Processor to communicate with the bank server.
- **Database Interface:** Allows authorized components to access account and transaction data.
- **Hardware Interface:** Allows the ATM software to interact with ATM hardware devices.

Provided interfaces are represented using the UML lollipop notation, while required interfaces are represented using the socket notation.

4. Data Flow and Interaction

First, the customer interacts with the ATM UI using the card reader and keypad. The Authentication Service validates the customer's credentials. After successful authentication, the Transaction Processor handles the selected transaction. It communicates with the Bank Server API when bank-side verification or transaction processing is required. The Database supports the storage and retrieval of account and transaction information. Finally, the ATM Hardware Adapter coordinates physical operations such as dispensing cash and printing receipts.

5. Conclusion

The component diagram provides a clear view of the ATM system's software structure and component interactions. Separating the system into independent components improves modularity, maintainability, testing, and future upgrades. The diagram also shows how the software communicates with external bank services and physical ATM devices.